

Luke Skywalker gave Chewbacca an integer number x. Chewbacca isn't good at numbers but he loves inverting digits in them. Inverting digit t means replacing it with digit 9 - t.

Help Chewbacca to transform the initial number x to the minimum possible positive number by inverting some (possibly, zero) digits. The decimal representation of the final number shouldn't start with a zero.

Input Format

The first line contains a single integer x (1 ≤ x ≤ 10^18) – the number that Luke Skywalker gave to Chewbacca.

Constraints

x ≤ 1000000000000000000

Output Format

Print the minimum possible positive number that Chewbacca can obtain after inverting some digits. The number shouldn't contain leading zeroes.

Sample Input

4545

Sample Output

4444

1 x 10 + 2

num = 1

2 1

94378

sum = 1

= 1 + 2 x 10

= 21 + 3 x 10^2

= 321 x 10^4

= 4321 x 10^4

9 x 10^4 + 4 x 10^3 + 3 x 10^2 + 7 x 10^1 + 8 x 10^0

943

94

9

9 4 3 7 8
↓ ↓ ↓ ↓ ↓
9 4 3 2 1

mul = 1 num = 94378

while(n > 0) {
 10^1 10^2 10^3 10^4

rem = n % 10

if (rem > 5) {

rem = 9 - rem

sum = sum + rem * mul

n = n / 10

mul = mul * 10

Faculty at CodingBlocks loves to purchase smartphones and decides to play a game. Aayush and Harshit decides to shop for smartphones. Aayush purchases 1 smartphone, then Harshit purchases 2 smartphones, then Aayush purchases 3 smartphones, then Harshit purchases 4 smartphones, and so on. Once someone can't purchase more smartphones, he loses.

Aayush can purchase at most M smartphones and Harshit can purchase at most N smartphones. Who will win ? Print "Aayush" and "Harshit" accordingly.

Input Format

The first line of the input contains an integer T denoting the number of test cases. The description of T test cases follows. Two integers M and N denoting the maximum possible number of smartphones Aayush and Harshit can purchase respectively.

Constraints

1 ≤ T ≤ 1000 1 ≤ M, N ≤ 10^6

Output Format

For each test case, output a single line containing one string – the name of the winner i.e. Aayush or Harshit

Sample Input

2
9 3
8 11

Sample Output

Aayush
Harshit

A = 0 , 40
phone = 1

A = 1

H = 2

A
M = 13

H
N = 15

while () {

A = A + phone

if (A > M) {

System.out.println("Aayush")

phone++

H = H + phone

if (H > N) {

System.out.println("Harshit")

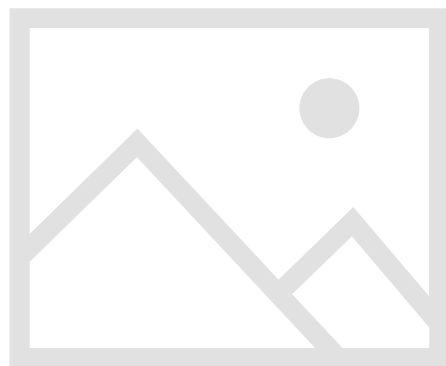
phone++



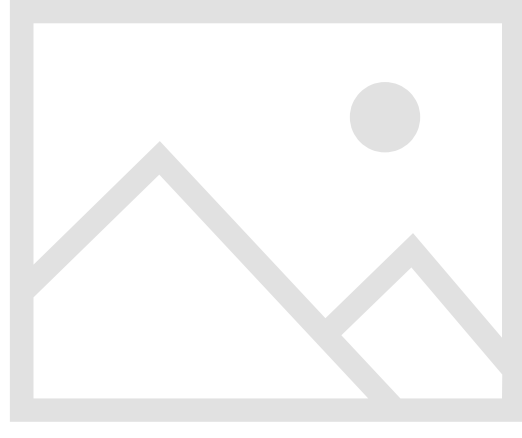
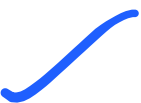
sb
8

db
2

sn
23



x



x

int sum = 0

while (true) {

a = s.nextInt()

sum = sum + a

if (sum < 0) {

break

System.out.println(a)